

Hybrid Machine-Learning / Molecular-Mechanics

Energy Functions for Condensed-Phase

Simulations: Training and Molecular Dynamics

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Abstract

We describe the hybrid machine-learning / molecular-mechanics (ML/MM) energy function implemented in the MMML software stack and the associated training and molecular-dynamics (MD) workflows. Intramolecular and short-range intermolecular interactions are evaluated with a neural-network potential, while intermolecular Lennard-Jones and Coulomb interactions are supplied by a CHARMM/CGenFF-compatible classical term that is switched on as a function of monomer center-of-mass separation. The same hybrid total energy is assembled during supervised training (`physnet-train`) and during condensed-phase MD (`md-system`), so that the learned model is trained on the Hamiltonian later used for dynamics. Mechanical and electrostatic embeddings for the classical Coulomb term are controlled by a charge-mode switch (`fixed` versus Q^0). Periodic long-range electrostatics are provided by optional Ewald / particle-mesh Ewald solvers. The Methods section below documents the energy decomposition, CLI entry points, default handoff cutoffs, and campaign pipelines used for liquid dichloromethane, acetone, and methane.

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1 Introduction

Machine-learned (ML) potentials offer near-quantum accuracy for local chemical environments, but long-range electrostatics and large condensed-phase systems remain expensive when treated entirely by neural networks. Classical molecular mechanics (MM) force fields such as CHARMM/CGenFF^{1,2} remain efficient for intermolecular interactions at longer range. Hybrid ML/MM energy functions combine the two: ML for intramolecular and short-range many-body interactions, and MM for the intermolecular tail and optional long-range Coulomb solvers.^{3,4}

A practical hybrid scheme must expose a single total energy both when fitting parameters and when integrating equations of motion. The MMML package implements that requirement through a shared hybrid forward pass used by the `physnet-train` and `md-system` commands. The remainder of this article details the hybrid energy (Section 2), training (Section 3), MD campaigns (Section 4), charge embeddings (Section 5), and software reproducibility (Section 6).

2 Hybrid ML/MM Energy Function

2.1 Monomer Partition and Center-of-Mass Switching

Interactions are partitioned by monomer (typically one solvent residue). Let r_{IJ} denote the center-of-mass (COM) distance between monomers I and J . The hybrid potential takes the form

$$E = \sum_I E_I^{\text{ML}} + \sum_{I < J} [s_{\text{ML}}(r_{IJ}) E_{IJ}^{\text{ML}} + s_{\text{MM}}(r_{IJ}) E_{IJ}^{\text{MM}}] + E^{\text{LR}} + E^{\text{wall/restr.}} \quad (1)$$

where E_I^{ML} is the intramolecular ML energy of monomer I , E_{IJ}^{ML} is the ML two-body (dimer) correction, and E_{IJ}^{MM} collects switched intermolecular Lennard-Jones and Coulomb contributions evaluated with CGenFF-compatible parameters. Optional long-range Coulomb beyond the real-space switch enters as E^{LR} (Ewald,⁵ particle-mesh Ewald, or related solvers). A short-range steric wall and restraints may contribute $E^{\text{wall/restr}}$.

For an isolated dimer AB , Equation 1 is equivalent to the training assembly

$$E = (1 - s)(E_A + E_B) + s E_{AB} + E_{\text{MM}}, \quad (2)$$

with $s = s_{\text{ML}}(r_{\text{COM}})$, $\Delta E_{\text{ML}} = E_{AB} - E_A - E_B$, and $E = E_A + E_B + s \Delta E_{\text{ML}} + E_{\text{MM}}$. The taper multiplies only the dimer interaction; monomer intramolecular energies remain fully on at all separations. Forces include the product-rule term arising from $\partial s / \partial \mathbf{R}$.

2.2 Default Handoff Window

With complementary handoff, $s_{\text{ML}} + s_{\text{MM}} = 1$ inside the $\text{ML} \leftrightarrow \text{MM}$ transition. Production defaults used throughout training and MD YAML are listed in Table 1 and illustrated in Figure 1.

Table 1: Default COM handoff parameters shared by `physnet-train` and `md-system`.

Parameter	Symbol / role	Default
<code>mm_switch_on</code>	end of $\text{ML} \rightarrow \text{MM}$ handoff	8.0 Å
<code>ml_switch_width</code>	ML taper width	1.5 Å
<code>mm_switch_width</code>	MM outer tail	5.0 Å

2.3 Ownership of Terms

Table 2 summarizes which component owns each contribution. Nuclear repulsion inside the neural model (optional Ziegler–Biersack–Littmark, ZBL) is distinct from the classical short-range wall used as a safety term outside the training domain.

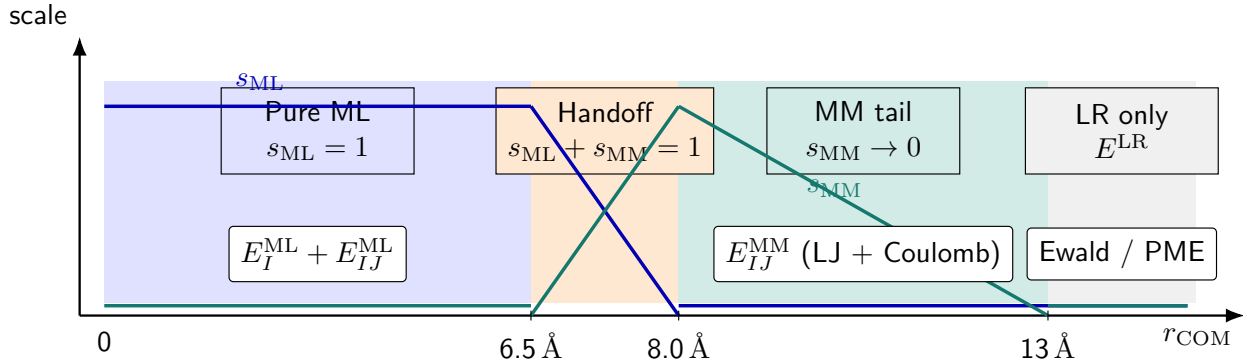


Figure 1: Schematic COM handoff for the hybrid energy in Equation 1. Below ≈ 6.5 Å the intermolecular ML dimer term is fully on; between 6.5 Å and 8.0 Å ML and MM scales are complementary; the switched MM pair term decays by 13 Å. Beyond the MM cutoff, only optional long-range Coulomb (E^{LR}) remains.

Table 2: Term ownership in the hybrid Hamiltonian.

Term	Owner	Notes
E_I^{ML}	PhysNet / Spooky	Always on
E_{IJ}^{ML}	PhysNet / Spooky	COM-gated by s_{ML}
LJ + Coulomb pairs	CGenFF MM	Scaled by s_{MM} ; q from charge mode
Optional LJ scales	Trainable s^σ, s^ϵ	Sidecar <code>hybrid_mm.json</code>
E^{LR}	Ewald / PME / ...	<code>lr_solver</code>
Bonded CHARMM	Topology (Mode B)	Or jax-mm-spoof bonded clone

3 Hybrid Training

3.1 CLI Entry Point and Dataset

Hybrid models are trained with

```
mmml physnet-train --hybrid-mm [--config train.yaml]
```

The CLI is implemented in `mmml/cli/make/make_training.py`. Training targets the hybrid total of Equation 2, not the bare neural energy, so that the checkpoint used later in MD matches the dynamics Hamiltonian.

Datasets are covalent dimers: each frame contains exactly two monomers. In addition to standard QM labels (R , Z , E , $\mathbf{F}$, optional dipole/charges), NPZ files must provide CGenFF

fields (`cgenff_type_idx`, `mol_id`, `cgenff_charge`, `cgenff_master_sigmas/epsilons`). These fields are prepared by `mmml prepare-mm-dataset` after electronic-structure evaluation of normal-mode samples.

3.2 Loss and Key Flags

The supervised objective is a weighted sum of squared residuals on hybrid energies and forces (and optional dipole / charge heads),

$$\mathcal{L} = w_E \|E - E^{\text{ref}}\|^2 + w_F \|\mathbf{F} - \mathbf{F}^{\text{ref}}\|^2 + w_D \|\boldsymbol{\mu} - \boldsymbol{\mu}^{\text{ref}}\|^2 + w_q \|q - q^{\text{ref}}\|^2. \quad (3)$$

Representative weights used in the bundled fixed-charge template are $w_E = 1$, $w_F = 52.91$, $w_D = 27.21$, and $w_q = 14.39$. Validation scores the same hybrid total.

Essential flags (YAML or CLI) are summarized below. The PhysNet radial cutoff must be at least `mm_switch_on`. Portable JSON checkpoints (and optional `hybrid_mm.json` LJ-scale sidecars) are written for MD deployment. Figure 2 summarizes the pipeline.

- `--hybrid-mm` — enable hybrid total of Equation 2
- `--mm-charge-mode` `{fixed,q0,latent,...}`
- `--charges` / `--no-charges` — neural charge head
- `--ml-switch-width`, `--mm-switch-on`, `--mm-switch-width` (Table 1)
- `--lr-solver` `{mic,ewald,nvalchemiops_pme}`
- `--learn-mm-lj-scales` — optional trainable LJ scales

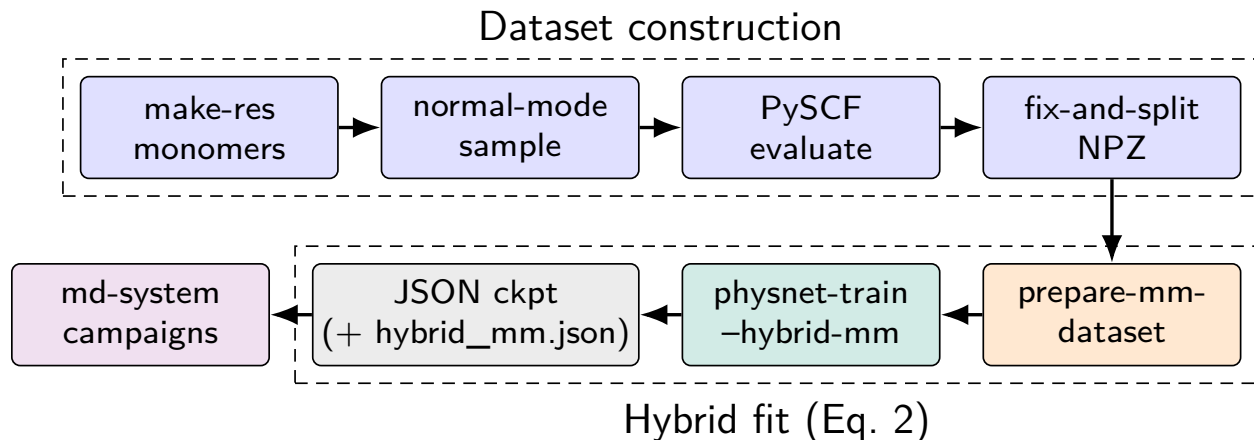


Figure 2: Hybrid training pipeline. Quantum labels on dimer geometries are augmented with CGenFF arrays, then `physnet-train --hybrid-mm` fits the hybrid total of Equation 2. The resulting portable JSON checkpoint is consumed unchanged by `md-system`.

4 Condensed-Phase Molecular Dynamics

4.1 CLI Entry Point and Campaign YAML

Dynamics are launched with

```
mmml md-system --config campaign.yaml --run-all [--resume]
```

(`mmml/cli/run/md_system.py`). A campaign file supplies global defaults (composition, box, checkpoint, cutoffs, `lr_solver`, `mm_charge_mode`, backend options) and an ordered list of runs/jobs. Dependencies are resolved automatically; each leg writes a handoff state (`handoff/state.npz`) consumed by the next stage.

Supported backends are `pycharm` (CHARMM integrator with MLPot USER callback),² `jaxmd` (pure-JAX evaluation with a jax-md integrator),⁶ and `ase` for reference smokes. Ensembles include `pbm_nvt`, `pbm_npt`, and `pbm_nve`.

4.2 Two Drivers, One Hamiltonian

- Mode A (jax-md). CHARMM is used only to build the topology; every energy term is evaluated in a jitted hybrid energy function.

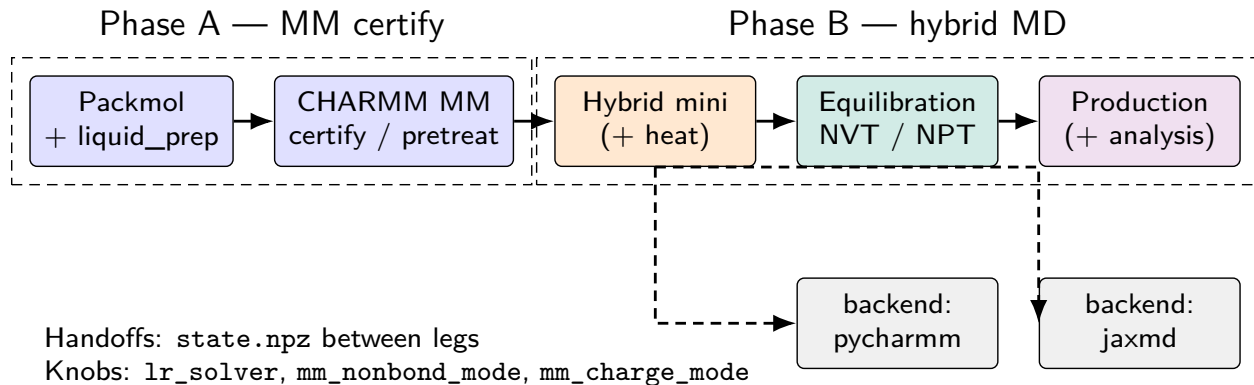


Figure 3: Condensed-phase campaign pipeline driven by `md-system --run-all`. Phase A builds and certifies a classical liquid box; Phase B registers the hybrid potential and runs mini/heat/equilibration/production on either the PyCHARMM or jax-md backend. Long-range Coulomb and charge embedding are selected in campaign defaults and applied identically on both backends.

- Mode B (PyCHARMM). CHARMM advances the trajectory; the hybrid ML contribution enters each step through the MLpot callback while classical bonded / nonbonded bookkeeping follows the PSF.

Both drivers evaluate Equation 1 with the same handoff parameters and charge mode.

4.3 Box Preparation and Dynamics Stages

Dense liquids follow a two-phase protocol (Figure 3):

1. Phase A (MM certify). Packmol placement, optional Monte Carlo density moves, and classical minimization / short dynamics until inter-monomer contact criteria pass.
2. Phase B (hybrid MD). Register the ML potential (or a jax-mm-spoof bonded clone for infrastructure tests), minimize under the hybrid energy, then heat $\rightarrow$ equilibrate $\rightarrow$ produce.

Optional CHARMM MM pretreat (`pretreat_on_pycharm`) inserts short classical heat/equilibration before MLpot registration on dense boxes.

4.4 Paper Workflows

Two Snakemake workflows instantiate the campaign pattern for the manuscript:

- `workflows/pbc_liquid_density_dyn/` — dichloromethane and acetone liquids under NPT/NVT with density acceptance metrics;
- `workflows/pbc_methane_ewald/` — methane at fixed liquid density with `lr_solver: ewald` and `mm_nonbond_mode: periodic_external`, sweeping temperature, checkpoint, backend, and embedding (mechanical / electrostatic).

5 Mechanical and Electrostatic Embeddings

The classical Coulomb term inside E_{MM} obtains its per-atom charges from `mm_charge_mode`. This choice is orthogonal to whether the neural model predicts charges for use inside E^{ML} .

Mechanical embedding (**fixed**). $q_{\text{MM}} = q_{\text{CGenFF}}$. Default for charge-less PhysNet checkpoints (e.g., DES dimers) and for LJ-fitting stages.

Electrostatic embedding (**q0 / Q⁰**). $q_{\text{MM}} = \text{neutralize}(Q^0)$, where Q^0 denotes unperturbed monomer charge-head evaluations. Requires a charge-capable checkpoint (`charges: true`). Suitable for liquids.

Dimer-only modes (**latent/Q¹**, **fixed_plus_latent**). Partner-perturbed charges from the AB forward; undefined for $N > 2$ monomers and therefore not used in liquid production matrices.

Figure 4 contrasts the two liquid-capable embeddings used in the methane Ewald smoke matrix. Campaign tags encode the choice as `_mech_` or `_es_`.

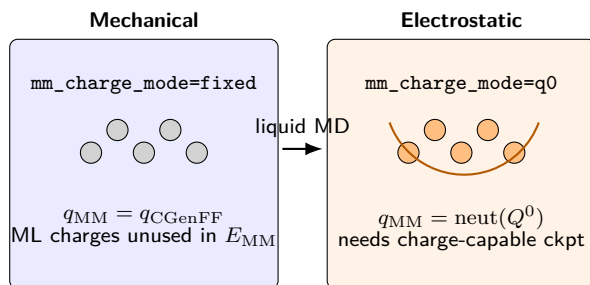


Figure 4: Liquid-capable embeddings for hybrid MM Coulomb. Mechanical embedding keeps fixed CGenFF charges in E_{MM} . Electrostatic embedding (Q^0) rebuilds neutralized monomer ML charges each evaluation. Dimer-only Q^1 modes are omitted from condensed-phase matrices.

6 Software and Reproducibility

Table 3 lists the principal components. Manuscript figures and tables are regenerable from versioned Snakemake workflows under `workflows/`; each Results entry should cite the workflow directory, config file, artifact path, and git commit. Engineering smoke matrices (including DES / So3LR methane Ewald cells with mechanical and electrostatic embeddings) are recorded in the accompanying Results draft and are not themselves production property claims.

Table 3: Software stack for hybrid training and MD.

Component	Role
MMML (<code>mmml</code>) ⁴	CLI, hybrid calculator, workflows
<code>physnet-train</code>	Hybrid supervised training
<code>md-system</code>	Campaign MD (ASE / jax-md / PyCHARMM)
JAX / jax-md ⁶	Differentiable hybrid energy; jaxmd backend
PyCHARMM / CHARMM ²	Topology, classical MM, pycharmm backend
Snakemake (+ Slurm)	Parameter sweeps and cluster submission

7 Conclusions

The hybrid ML/MM formulation in Equation 1, the training assembly in Equation 2, and the `md-system` campaign protocol provide a closed loop from dimer QM data to condensed-phase dynamics. Mechanical (`fixed`) and electrostatic (Q^0) embeddings select how classical Coulomb charges are obtained without changing the ML/MM ownership of bonded and short-range many-body terms. Shared handoff defaults and portable JSON checkpoints keep training and MD numerically aligned across the `jax-md` and `PyCHARMM` drivers.

Acknowledgement

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Supporting Information Available

Extended CLI flag tables, YAML templates under `examples/hybrid_mm_charges/`, and workflow provenance maps are available in the MMML documentation tree `docs/manuscripts/condensed-`

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